Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Calcalkaline**
Data suite	**./adamello.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Primitive Basic Intermediate Evolved

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 70.80 wt.%

	Primitive	Basic	Intermediate	Evolved	Total
Olivine	9.01				
Spinel	11.81				
Clinopyroxene	70.81				
Orthopyroxene		3.62	6.94		4.39
Plagioclase	8.37	32.26	67.98	65.91	50.54
Amphibole		49.6	17.52	18.07	33.0
Biotite				0.05	0.01
Quartz				12.96	1.78
Magnetite		14.42	6.87	2.01	9.83
Apatite		0.1	0.7	1.0	0.45
FC%	32.0	34.0	41.0	25.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	3.2	3.2	3.2	3.2	3.2
Clinopyroxene	0.28	0.28	0.28	0.28	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Primary	Primitive	Basic	Intermediate	Evolved	End
Sample	RN6V2	RN8	RN10	RN15V2	RN17V2	
SiO2	49.87	52.59	60.17	70.88	76.38	
TiO2	0.76	0.88	0.64	0.19	0.12	
Al2O3	16.41	19.72	19.52	16.33	13.75	
Fe2O3	9.52	9.6	4.86	1.56	0.79	
MgO	8.69	4.76	2.86	1.4	0.72	
CaO	12.07	8.4	6.92	2.76	0.58	
Na2O	2.02	3.04	3.65	4.76	4.99	
K2O	0.65	1.02	1.38	2.11	2.66	
P	480.26	698.56	960.52	698.56	174.64	
Ba	0.0	0.0	0.0	0.0	0.0	
Rb	0.0	0.0	0.0	0.0	0.0	
Sr	0.0	0.0	0.0	0.0	0.0	
Nb	0.0	0.0	0.0	0.0	0.0	
Zr	0.0	0.0	0.0	0.0	0.0	
Y	0.0	0.0	0.0	0.0	0.0	
La	0.0	0.0	0.0	0.0	0.0	
Ce	0.0	0.0	0.0	0.0	0.0	

	Primary	Primitive	Basic	Intermediate	Evolved	End
Nd		0.0	0.0	0.0	0.0	0.0
Sm		0.0	0.0	0.0	0.0	0.0
Eu		0.0	0.0	0.0	0.0	0.0
Gd		0.0	0.0	0.0	0.0	0.0
Tb		0.0	0.0	0.0	0.0	0.0
Dy		0.0	0.0	0.0	0.0	0.0
Ho		0.0	0.0	0.0	0.0	0.0
Er		0.0	0.0	0.0	0.0	0.0
Tm		0.0	0.0	0.0	0.0	0.0
Yb		0.0	0.0	0.0	0.0	0.0
Lu		0.0	0.0	0.0	0.0	0.0
Hf		0.0	0.0	0.0	0.0	0.0
Ta		0.0	0.0	0.0	0.0	0.0
Th		0.0	0.0	0.0	0.0	0.0
U		0.0	0.0	0.0	0.0	0.0

Table 3: Major oxide compositions of the model phases (wt.% for oxides, ppm for trace elements).

Computed Compositions

	Primary	Primitive	Basic	Intermediate	Evolved	End
SiO2		49.87	52.59	60.54	72.23	75.8
TiO2		0.76	0.88	0.52	-0.02	-0.06
Al2O3		16.41	19.72	19.21	15.22	13.68
Fe2O3		9.52	9.6	4.8	1.5	0.77
MgO		8.69	4.76	2.87	1.45	0.79
CaO		12.07	8.4	6.95	2.46	1.31
Na2O		2.02	3.04	3.72	5.14	5.29
K2O		0.65	1.02	1.41	2.27	2.92
P		480.26	698.56	957.21	666.68	234.01
Ba		0.0	0.0			
Rb		0.0	0.0			
Sr		0.0	0.0			
Nb		0.0	0.0			
Zr		0.0	0.0			
Y		0.0	0.0			
La		0.0	0.0			
Ce		0.0	0.0			
Nd		0.0	0.0			
Sm		0.0	0.0			
Eu		0.0	0.0			
Gd		0.0	0.0			
Tb		0.0	0.0			
Dy		0.0	0.0			
Ho		0.0	0.0			
Er		0.0	0.0			
Tm		0.0	0.0			

	Primary	Primitive	Basic	Intermediate	Evolved	End
Yb		0.0	0.0			
Lu		0.0	0.0			
Hf		0.0	0.0			
Ta		0.0	0.0			
Th		0.0	0.0			
U		0.0	0.0			

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primitive > Basic

PRIMITIVE

Population size	50
Iterations	500/500
Cost function	0.32
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	49.87	52.59	52.65	0.12
TiO2	0.76	0.88	0.96	9.27
Al2O3	16.41	19.72	19.78	0.3
Fe2O3	9.52	9.6	9.64	0.33
MgO	8.69	4.76	4.75	0.15
CaO	12.07	8.4	8.44	0.4
Na2O	2.02	3.04	2.85	6.34
K2O	0.65	1.02	0.94	7.11

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	32.0
Olivine	9.01
Spinel	11.81
Clinopyroxene	70.81
Orthopyroxene	0.0
Plagioclase	8.37
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0

	Value
Magnetite	0.0
Ilmenite	0.0

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	43.95	0.33	9.26	9.28	17.08	19.81	0.28	0.02

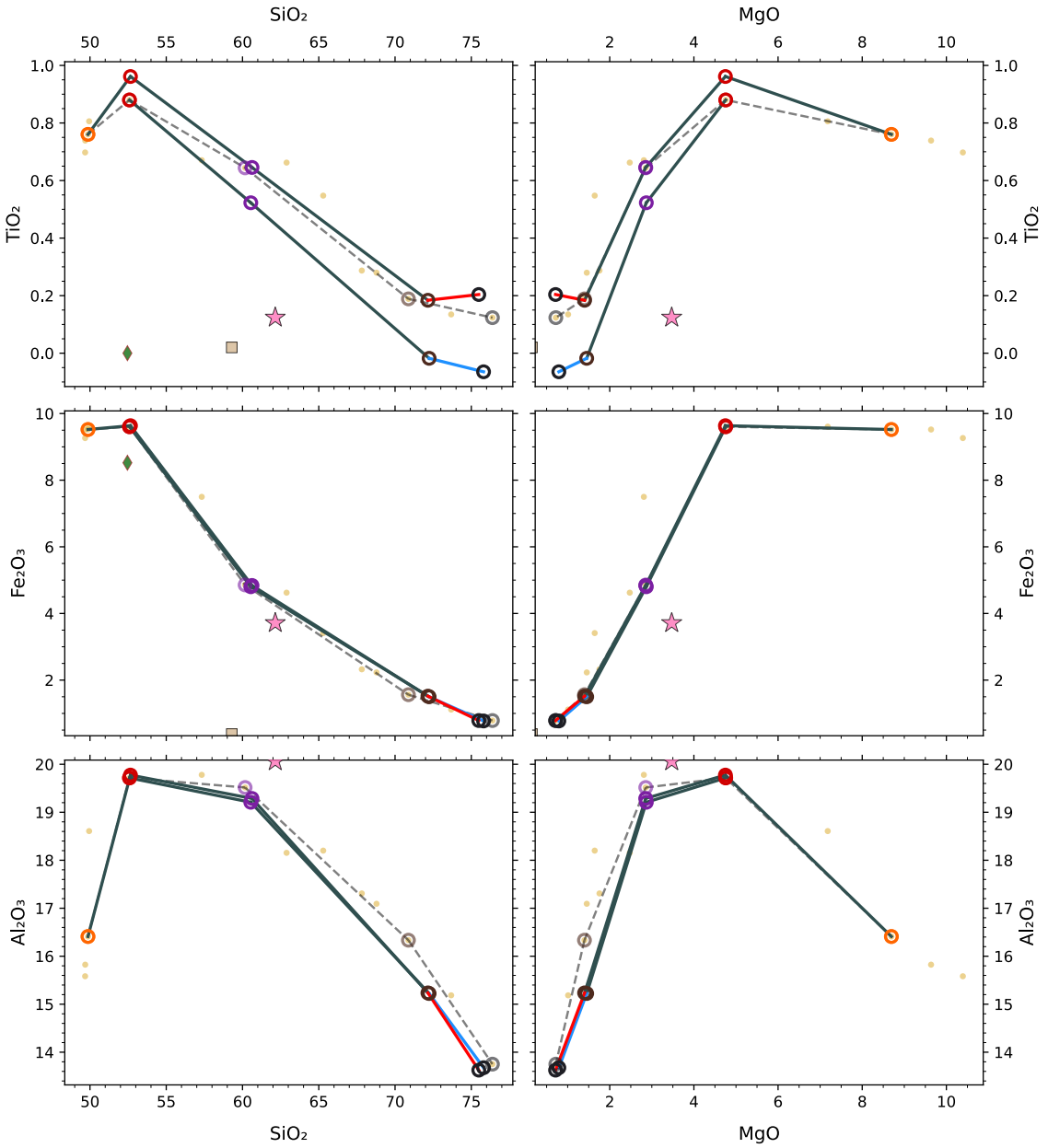
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	38.67	0.02	0.03	17.48	43.5	0.27	0.01	0.01
spinel	0.26	0.53	30.33	30.52	13.68	24.34	0.34	0.0
clinopyroxene	51.69	0.38	3.97	5.69	16.3	21.72	0.24	0.01
orthopyroxene								
plagioclase	45.77	0.0	34.23	0.83	0.0	18.24	0.82	0.11
amphibole								
kfeldspar								
biotite								
quartz								
magnetite								
ilmenite								

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

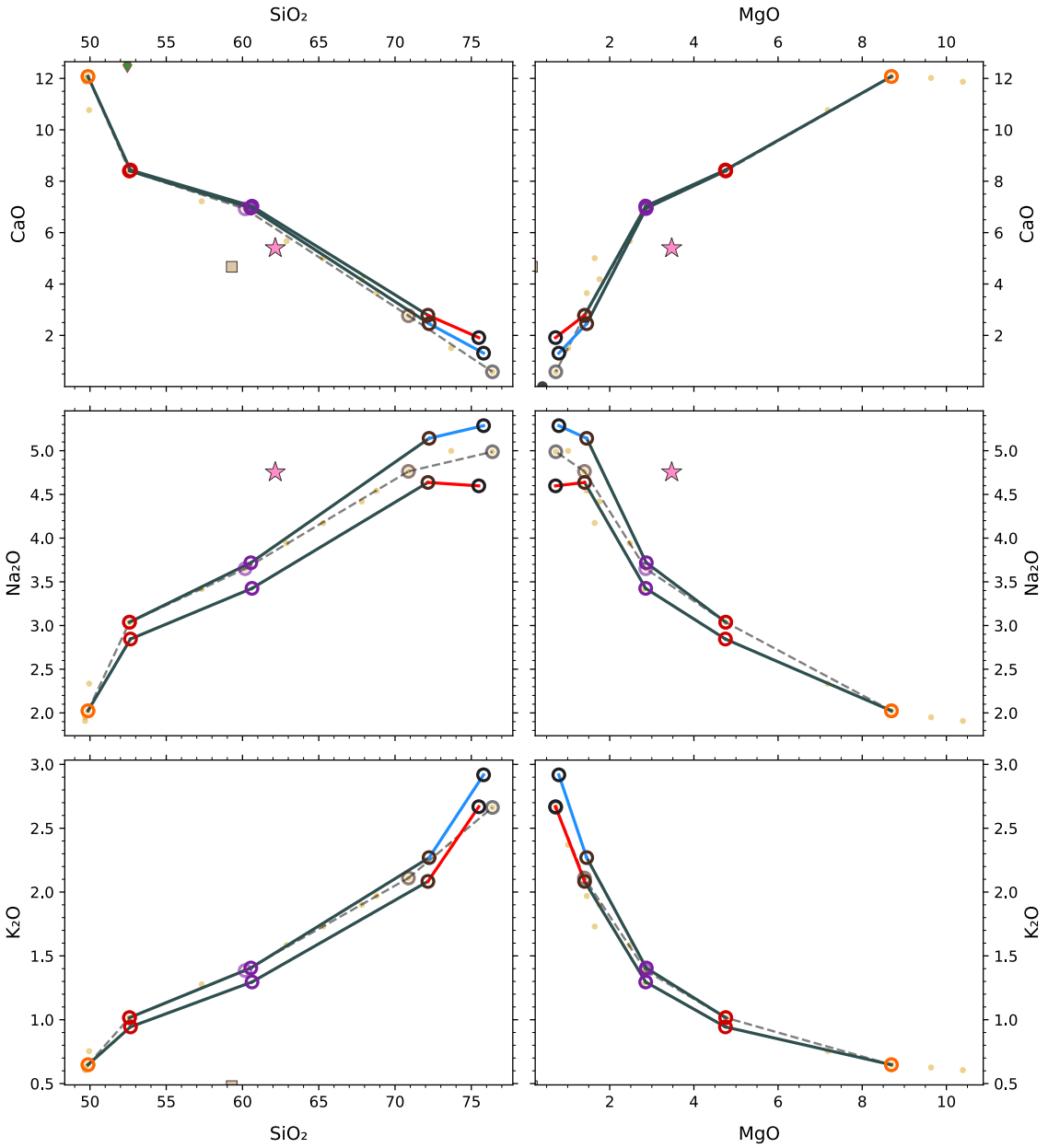


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	0.35
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	52.59	60.17	60.54	0.62
TiO2	0.88	0.64	0.52	18.83

	Initial	Expected	Computed	% Error
Al2O3	19.72	19.52	19.21	1.58
Fe2O3	9.6	4.86	4.8	1.11
MgO	4.76	2.86	2.87	0.49
CaO	8.4	6.92	6.95	0.44
Na2O	3.04	3.65	3.72	1.87
K2O	1.02	1.38	1.41	1.47
P	698.56	960.52	957.21	0.34
Ba	0.0	0.0		
Rb	0.0	0.0		
Sr	0.0	0.0		
Nb	0.0	0.0		
Zr	0.0	0.0		
Y	0.0	0.0		
La	0.0	0.0		
Ce	0.0	0.0		
Nd	0.0	0.0		
Sm	0.0	0.0		
Eu	0.0	0.0		
Gd	0.0	0.0		
Tb	0.0	0.0		
Dy	0.0	0.0		
Ho	0.0	0.0		
Er	0.0	0.0		
Tm	0.0	0.0		
Yb	0.0	0.0		
Lu	0.0	0.0		
Hf	0.0	0.0		
Ta	0.0	0.0		
Th	0.0	0.0		
U	0.0	0.0		

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	34.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	0.0
Orthopyroxene	3.62
Plagioclase	32.26
Amphibole	49.6
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	14.42

	Value
Ilmenite	0.0
Apatite	0.1
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
wt.%	37.15	1.57	20.7	18.92	8.42	11.21	1.72	0.26	196.47

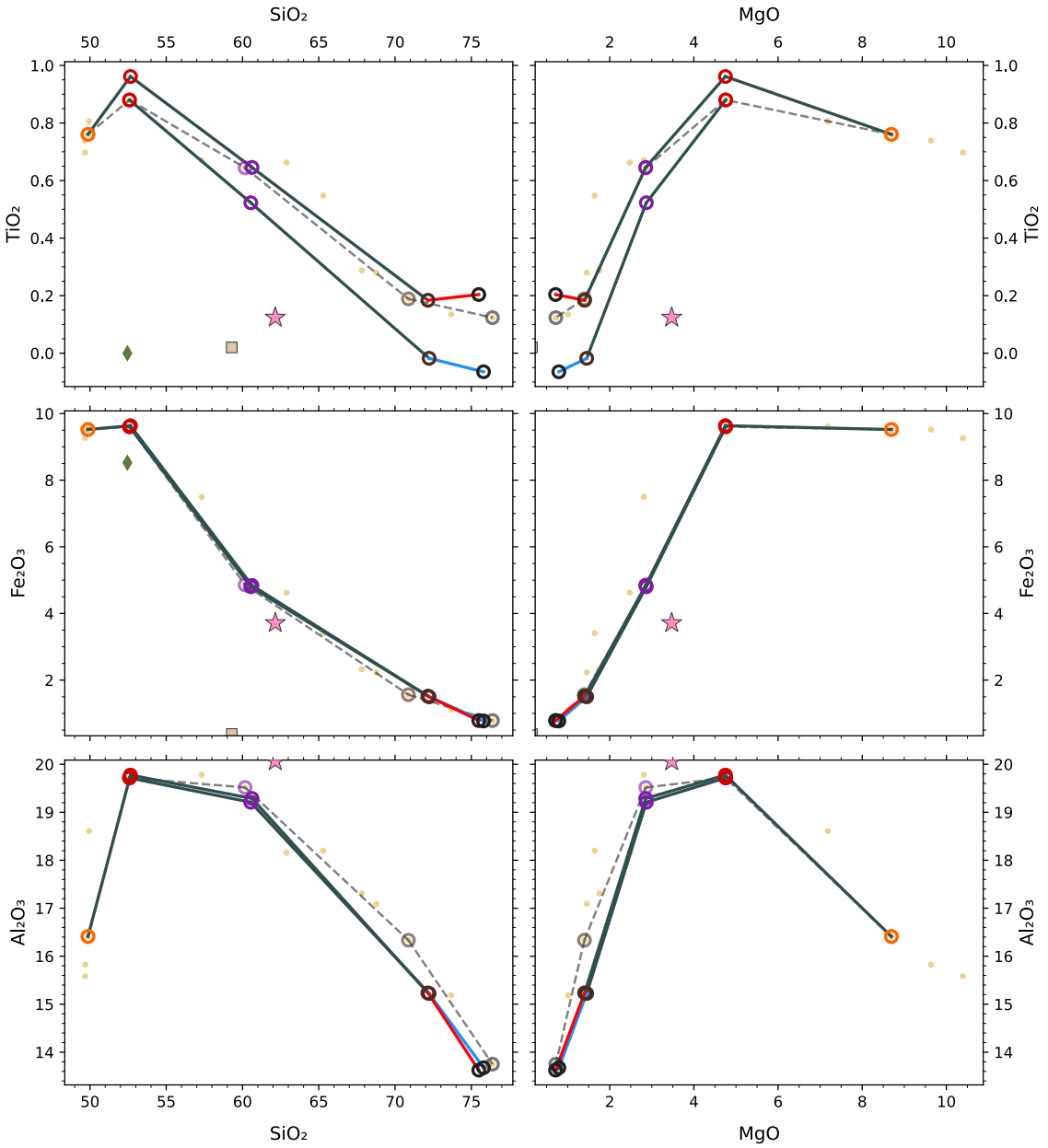
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
Olivine									
Spinel									
Clinopyroxene									
Orthopyroxene	49.34	0.37	6.38	17.8	24.43	1.64	0.03	0.02	0.0
Plagioclase	46.68	0.02	33.35	0.74	0.06	17.37	1.71	0.06	0.0
Amphibole	40.91	1.76	15.53	14.03	13.92	11.03	2.34	0.48	0.0
K-feldspar									
Biotite									
Quartz									
Magnetite	0.07	4.72	13.92	76.86	4.23	0.15	0.03	0.02	0.0
Ilmenite									
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0
Zircon									
Monazite									
Allanite									

Table 12: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

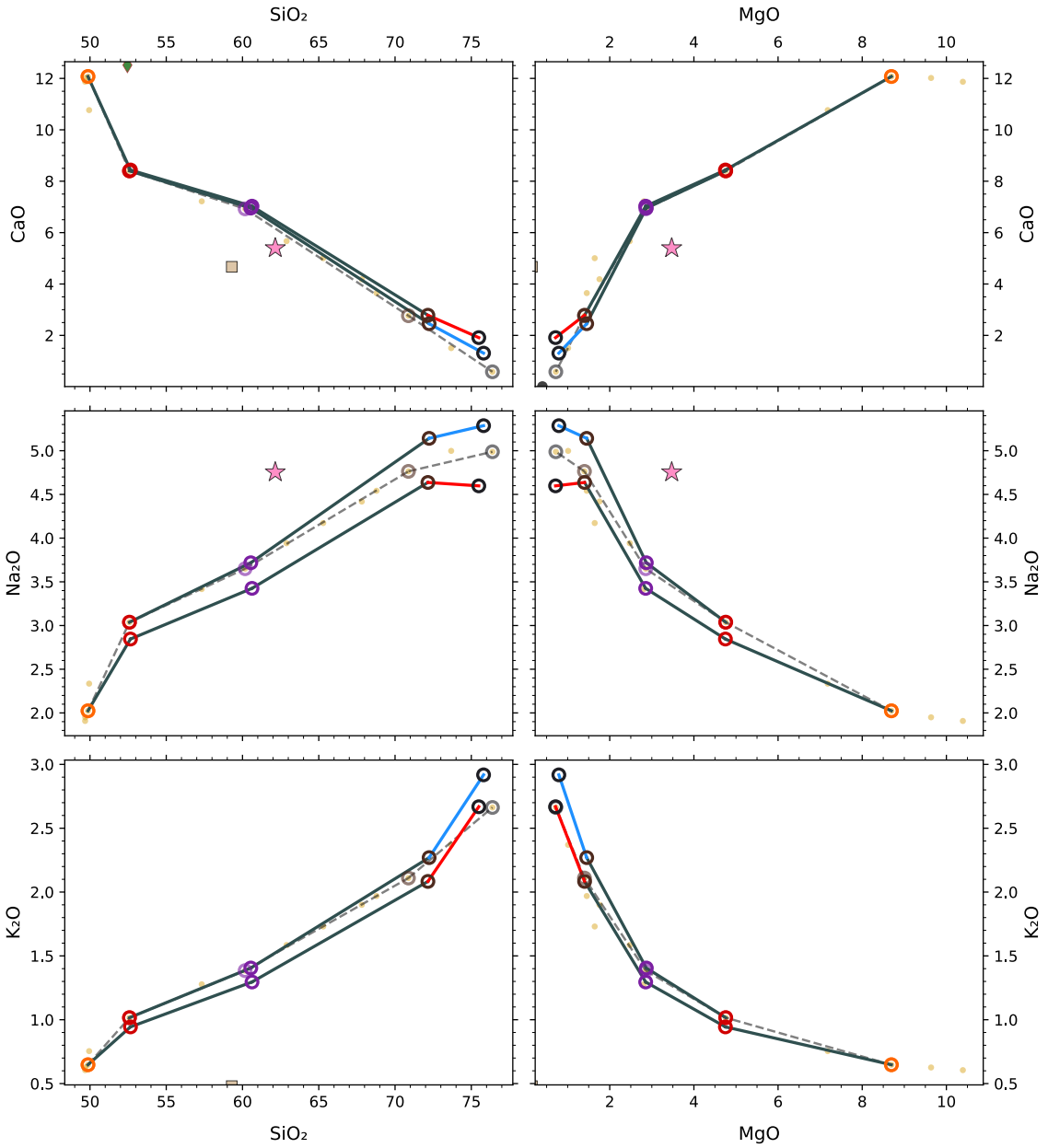


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Intermediate > Evolved

INTERMEDIATE

Population size	50
Iterations	500/500
Cost function	0.56
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	60.54	70.88	72.23	1.91
TiO2	0.52	0.19	-0.02	109.4

	Initial	Expected	Computed	% Error
Al2O3	19.21	16.33	15.22	6.8
Fe2O3	4.8	1.56	1.5	4.02
MgO	2.87	1.4	1.45	3.82
CaO	6.95	2.76	2.46	11.19
Na2O	3.72	4.76	5.14	7.93
K2O	1.41	2.11	2.27	7.58
P	957.21	698.56	666.68	4.56
Ba		0.0		
Rb		0.0		
Sr		0.0		
Nb		0.0		
Zr		0.0		
Y		0.0		
La		0.0		
Ce		0.0		
Nd		0.0		
Sm		0.0		
Eu		0.0		
Gd		0.0		
Tb		0.0		
Dy		0.0		
Ho		0.0		
Er		0.0		
Tm		0.0		
Yb		0.0		
Lu		0.0		
Hf		0.0		
Ta		0.0		
Th		0.0		
U		0.0		

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	41.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	0.0
Orthopyroxene	6.94
Plagioclase	67.98
Amphibole	17.52
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	6.87

	Value
Ilmenite	0.0
Apatite	0.7
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
wt.%	43.72	1.3	24.94	9.55	4.91	13.43	1.67	0.16	1375.29

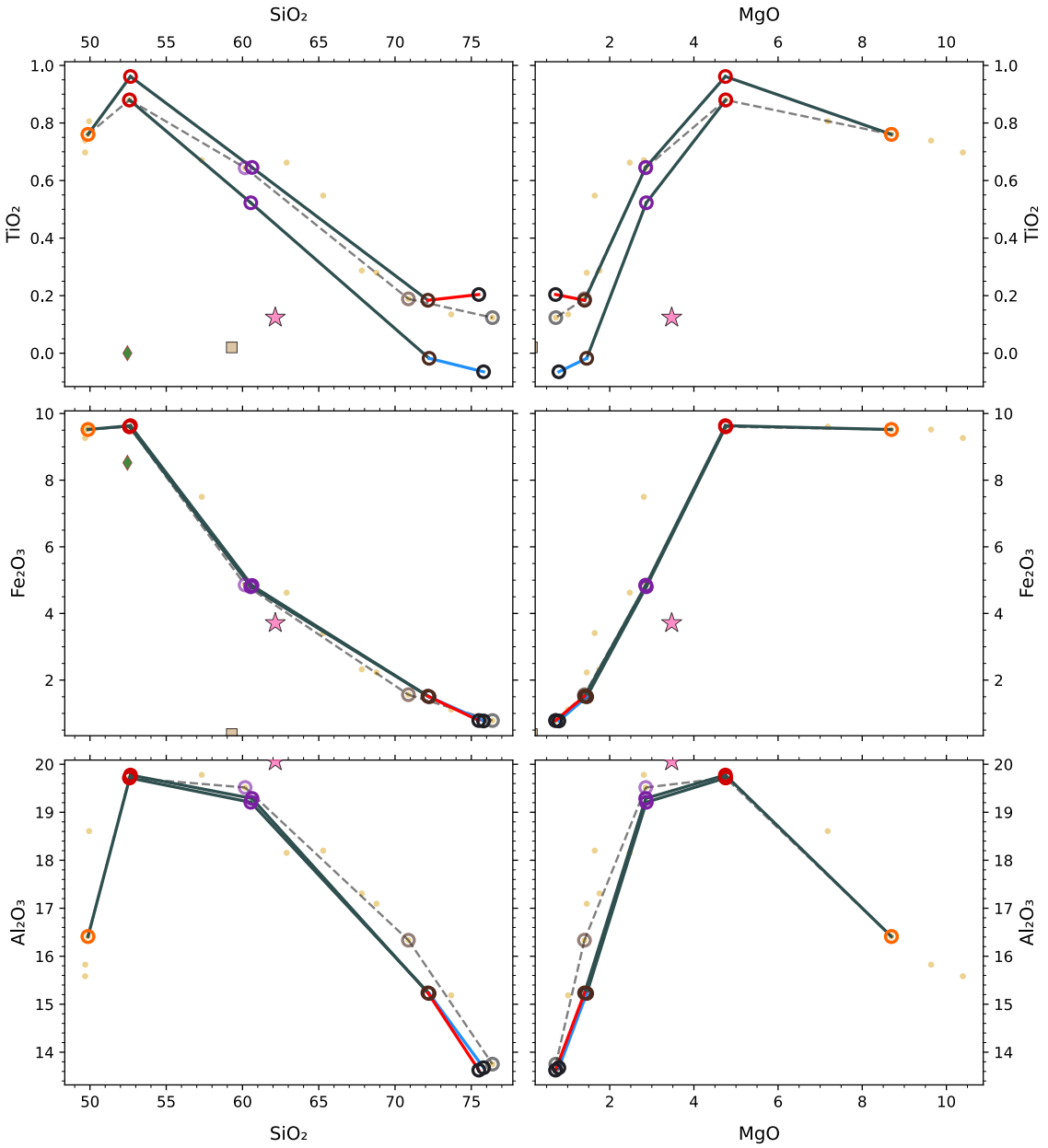
Table 15: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
Olivine									
Spinel									
Clinopyroxene									
Orthopyroxene	51.78	0.28	5.84	14.01	26.66	1.27	0.09	0.07	0.0
Plagioclase	47.78	0.06	32.59	1.06	0.33	16.18	1.88	0.12	0.0
Amphibole	43.62	2.01	13.26	11.58	15.8	11.14	2.19	0.41	0.0
K-feldspar									
Biotite									
Quartz									
Magnetite	0.13	12.93	0.89	84.91	0.99	0.08	0.03	0.04	0.0
Ilmenite									
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0
Zircon									
Monazite									
Allanite									

Table 16: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

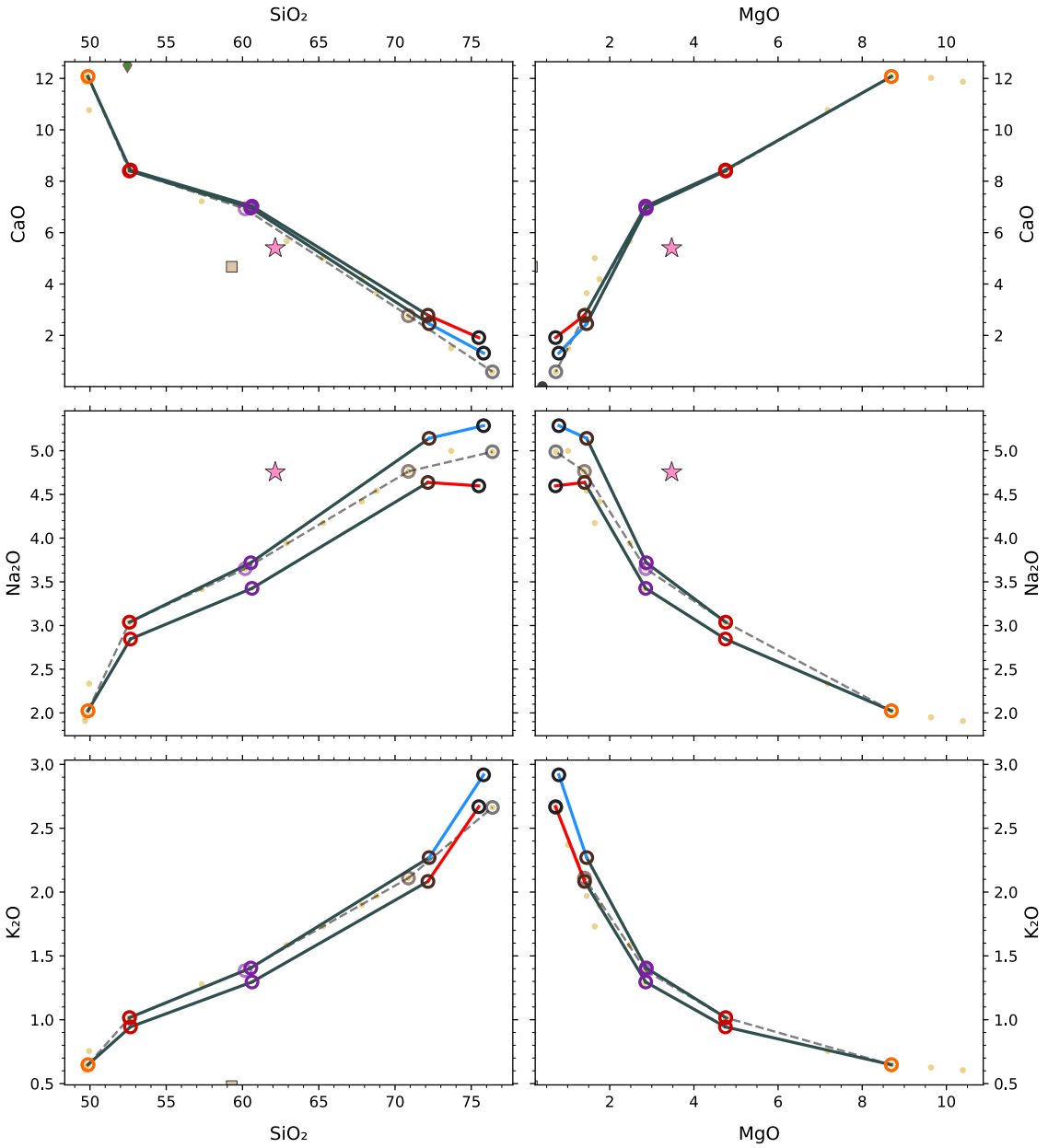


Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Evolved > End

EVOLVED

Population size	50
Iterations	500/500
Cost function	2.29
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	72.23	76.38	75.8	0.76
TiO2	-0.02	0.12	-0.06	152.25

	Initial	Expected	Computed	% Error
Al2O3	15.22	13.75	13.68	0.54
Fe2O3	1.5	0.79	0.77	1.61
MgO	1.45	0.72	0.79	9.97
CaO	2.46	0.58	1.31	124.14
Na2O	5.14	4.99	5.29	5.98
K2O	2.27	2.66	2.92	9.59
P	666.68	174.64	234.01	34.0
Ba		0.0		
Rb		0.0		
Sr		0.0		
Nb		0.0		
Zr		0.0		
Y		0.0		
La		0.0		
Ce		0.0		
Nd		0.0		
Sm		0.0		
Eu		0.0		
Gd		0.0		
Tb		0.0		
Dy		0.0		
Ho		0.0		
Er		0.0		
Tm		0.0		
Yb		0.0		
Lu		0.0		
Hf		0.0		
Ta		0.0		
Th		0.0		
U		0.0		

Table 17: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	25.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	0.0
Orthopyroxene	0.0
Plagioclase	65.91
Amphibole	18.07
K-feldspar	0.0
Biotite	0.05
Quartz	12.96
Magnetite	2.01

	Value
Ilmenite	0.0
Apatite	1.0
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 18: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
wt.%	61.53	0.12	19.86	3.68	3.44	5.89	4.71	0.32	1964.7

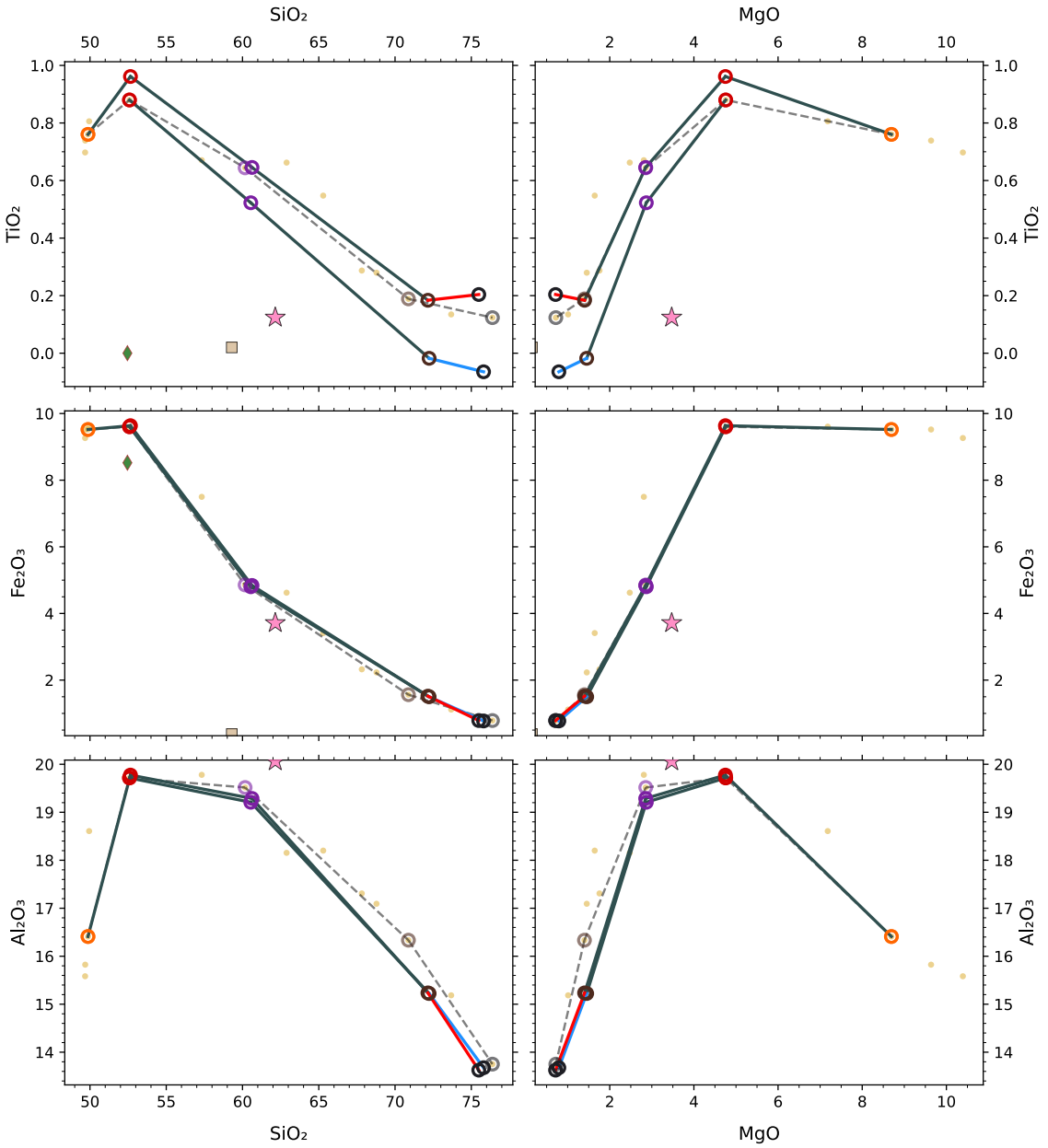
Table 19: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
Olivine									
Spinel									
Clinopyroxene									
Orthopyroxene									
Plagioclase	59.29	0.02	28.04	0.38	0.15	4.67	6.97	0.48	0.0
Amphibole	52.45	0.0	7.51	8.52	18.38	12.52	0.62	0.0	0.0
K-feldspar									
Biotite	41.37	2.07	17.6	9.43	19.77	0.01	0.92	8.82	0.0
Quartz	99.83	0.0	0.08	0.04	0.0	0.03	0.0	0.02	0.0
Magnetite	0.31	5.4	0.42	93.36	0.4	0.01	0.03	0.07	0.0
Ilmenite									
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0
Zircon									
Monazite									
Allanite									

Table 20: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

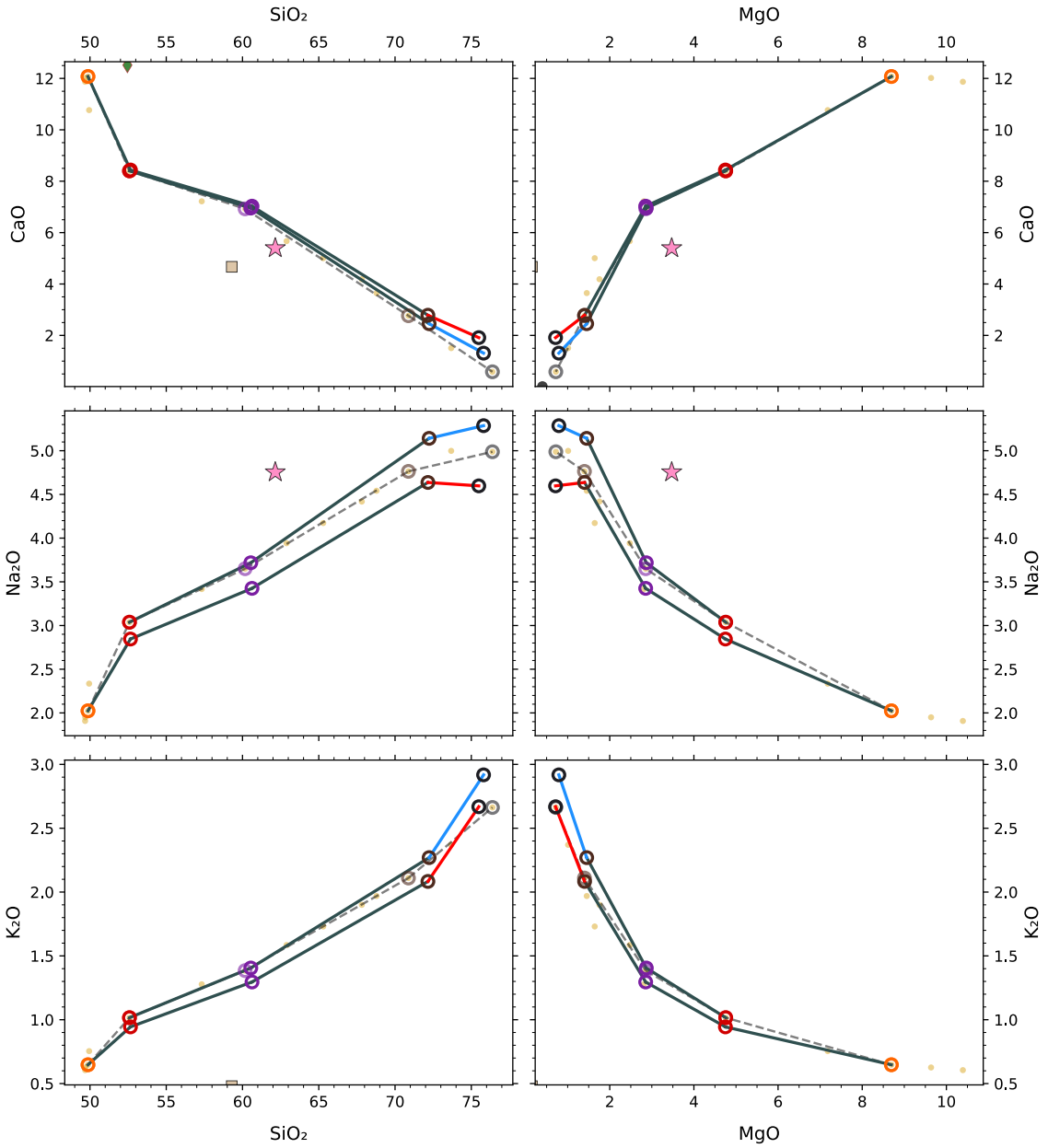


Figure 4: Bivariate plots of the data suite with the data-based and modelled LLODs. The Evolved to End transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

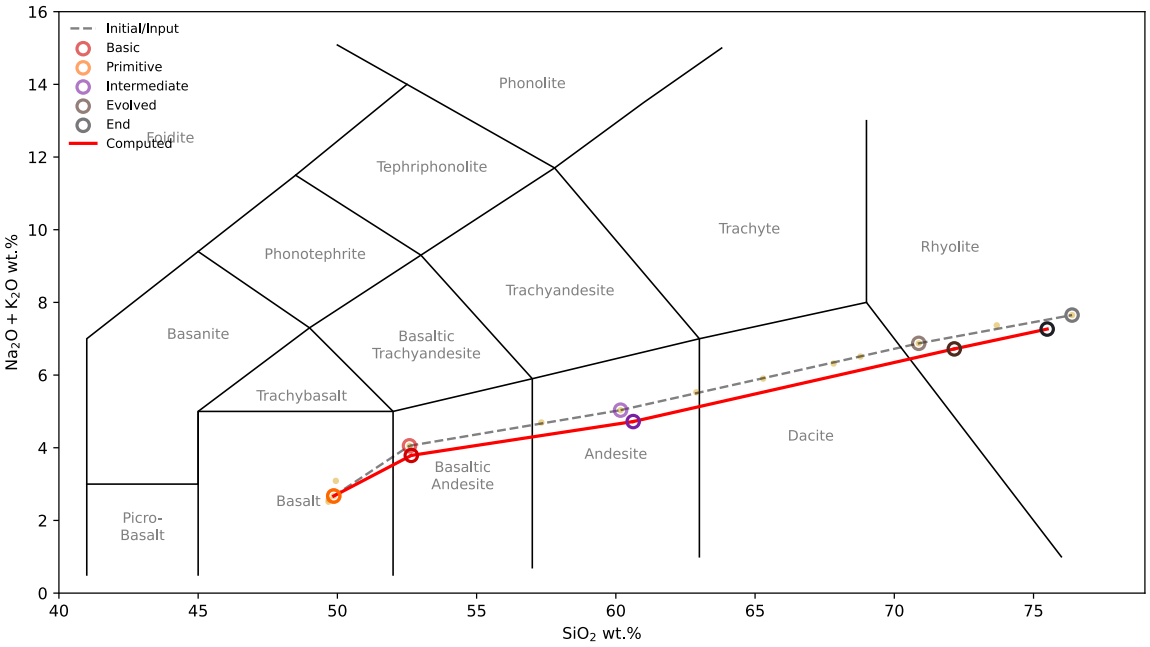


Figure 5: Total Alkali-Silica (TAS) diagram of the model phases.

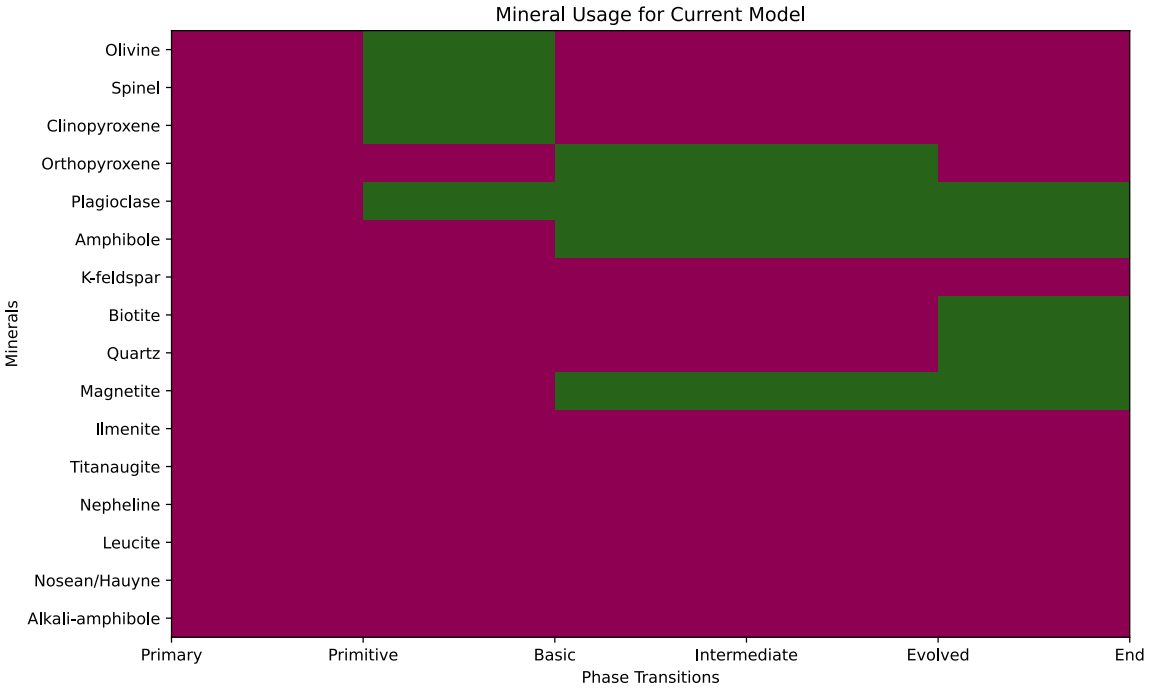


Figure 6: Mineral usage across the model phases.

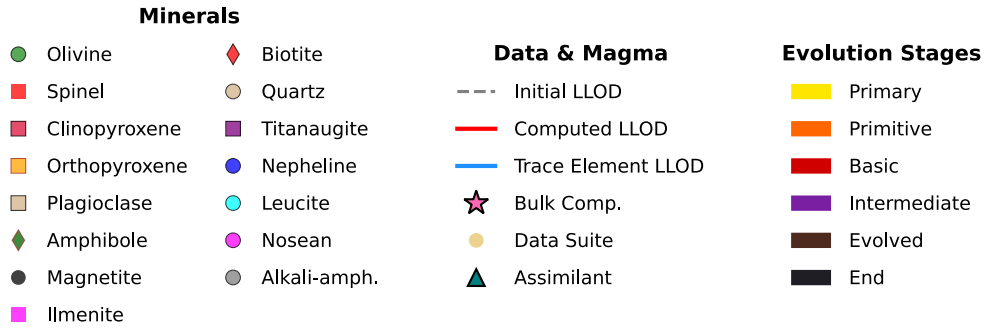


Figure 7: Legend for mineral symbols used in the figures.